

Homework — 1.3.1 Compression, Encryption and Hashing

Name: _____ Date: _ **Mark:** _____ / 20

OCR H446 — set after the 1.3.1 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. A backup program stores software installers and a folder of source-code text files. State which type of compression (**lossy** or **lossless**) must be used, and give **one** reason why this type is essential for these files. **[2]** (AO1/AO2)

2. Apply **run-length encoding (RLE)** to the following row of pixels. Write your answer as value–count pairs. **[2]** (AO2)

R R R R G G G R R B B B B

3. A different row of pixels is R G R G R G R G . Explain why applying RLE to this row is a poor choice, referring to the size of the encoded output. **[2]** (AO2)

4. Explain how **dictionary coding** reduces the size of a large text document, and state what must be available alongside the compressed data so the original document can be rebuilt. **[3]** (AO1/AO2)

5. Alice wants to send a confidential file to Bob, whom she has never met, over the internet. Explain how **asymmetric encryption** allows Bob to receive the file so that only he can read it. Refer to the public key and the private key in your answer. **[3]** (AO2)

6. A website stores a hash of each user's password rather than the password itself.

(a) State **one** property of a hash function that keeps the passwords safe even if the database of hashes is stolen. **[1]** (AO1)

(b) Describe how the system checks that a user has entered the correct password at login, without ever storing the plaintext password. **[1]** (AO2)

Part B — Retrieval: keep it fresh (~6 marks)

7. (1.2.2) Explain **one** difference between a **compiler** and an **interpreter**. **[2]** (AO1)

8. (1.1.1) State what is held in the **MAR** and the **MDR** during the fetch stage of the fetch–decode–execute cycle. **[2]** (AO1)

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9. (1.2.1) State the purpose of the **scheduler** in an operating system, and give **one** aim it tries to achieve.
[2] (AO1)

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